

Maincraft Technology

AIML Internship Task 1: House Price Prediction using Linear Regression

1. Introduction

The objective of this project is to build and evaluate a Linear Regression model using the California Housing dataset. The project demonstrates the complete machine learning workflow including data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and prediction.

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values based on relationships between variables.

2. Dataset Description

The California Housing dataset was obtained from the Scikit-Learn library. The dataset contains housing-related information collected from California districts.

Features Used

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

Target Variable

- House Price

The dataset contains multiple numerical features that help predict housing prices.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and relationships within the dataset.

Steps Performed

- Checked dataset shape and information
- Verified missing values
- Generated statistical summary

- Created visualizations

Visualizations Used

- Correlation Heatmap
- House Price Distribution
- Median Income vs House Price Scatter Plot
- Residual Distribution Plot

Key Observations

- Median Income showed strong positive correlation with house prices.
- Geographical features such as Latitude and Longitude also influenced pricing.
- No missing values were found in the dataset.
- The target variable showed moderate skewness.

4. Model Building

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

A Linear Regression model was trained using Scikit-Learn.

Linear Regression formula:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where:

- y = predicted house price
- x_1, x_2 = input features
- β = coefficients

The model learned relationships between housing features and house prices.

5. Model Evaluation

The model performance was evaluated using the following metrics:

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE measures the average prediction error.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE penalizes larger prediction errors.

R² Score

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

R² score indicates how well the model explains the variance in house prices.

Evaluation Results

- The model achieved good prediction accuracy.
- Predicted values were reasonably close to actual values.
- Residual errors were mostly normally distributed.

6. Results and Findings

The Linear Regression model successfully predicted house prices using housing-related features.

Important Findings

- Median Income was the most influential feature.
- Higher income areas generally had higher house prices.
- The model performed well for basic regression analysis.

Actual vs Predicted Analysis

The scatter plot between actual and predicted prices showed a positive linear relationship, indicating good model performance.

7. Future Improvements

The project can be improved using:

- i. Feature Engineering
- ii. Data Normalization
- iii. Hyperparameter Tuning
- iv. Advanced Algorithms such as:

- Random Forest Regression
- Decision Tree Regression
- XGBoost
- Neural Networks

Additional visualizations and larger datasets can further improve prediction performance.